Alternate Interior angles

Each of pair of angles named $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$ are marked on the opposite side of the transversal l and are lying between lines m and n are called alternate interior angles.

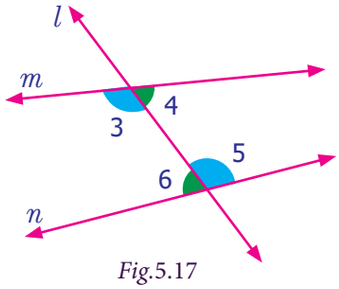


*Fig.*5.17

Alternate Exterior angles

Each pair of angles named $\angle 1$ and $\angle 7$, $\angle 2$ and $\angle 8$ are marked on the opposite side of the transversal l and are lying outside of the lines m and n are called alternate exterior angles.

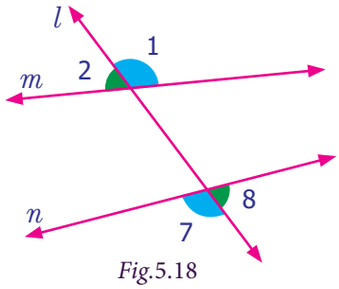


*Fig.*5.18

Now let us observe some more pairs of angles.

Each pair of angles named $\angle 3$ and $\angle 6$, $\angle 4$ and $\angle 5$ are marked on the same side of transversal l and are lying between the lines m and n. These angles are lying on the interior of the lines m and n as well as the same side of the transversal l.

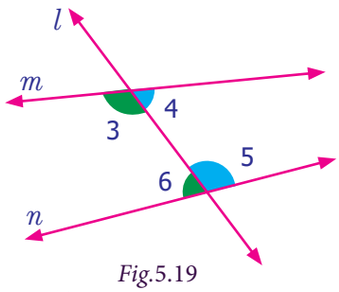


*Fig.*5.19

Each pair of angles named $\angle 1$ and $\angle 8$, $\angle 2$ and $\angle 7$ are marked on the same side of transversal l and are lying outside of the lines m and n. These angles are lying on the exterior of the lines m and n as well as the same side of the transversal l.

How can we call such angles?

We call them as co-interior and co-exterior angles.

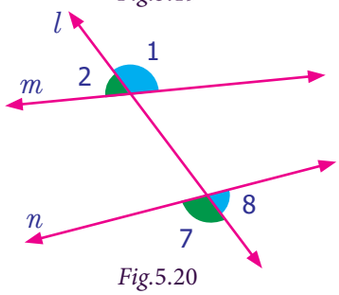


*Fig.*5.20



In Architecture, corresponding angles are used to assure symmetry and balance when designing the structure. Alternate exterior angles are used to ensure symmetry in floor plans. Alternate interior Angles are used to ensure that two beams are parallel and do not let the structure bend or deform in any form.

Interior angles on the same side of the transversal are used to determine if two beams are parallel and will not result in any distractions to the overall design. Exterior angles on the same side of the transversal are used to confirm that the walls are indeed straight and not at a different angle.

5.3.2 Angles formed by a transversal with Parallel lines

We saw different types of angles formed by a transversal while intersecting two given lines. Now let us observe some interesting facts on the angles formed by a transversal with the parallel lines from the following activities.

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